# Akoizo Original Practice - Form 1 - Answer key

### Division B | Scrambler | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

**1. (2 points) Which wins: a lower or higher score?**

Unit: Rules and the low-score system.
Worked-case data: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations. Find the Run Score.
Separate investigation observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

A. Lower

B. Higher

C. Closest to 100

D. Fewest runs

**Correct: A**

**2. (2 points) What is the Can Bonus for an Inside Can Distance of 40.0 cm?**

A. -55 points

B. -70 points

C. -35 points

D. -20 points

**Correct: C**

**3. (2 points) Which brake power source is allowed?**

Unit: Automatic stopping.
Worked-case data: Target 9.25 m. Wheels are 7.50 cm in diameter. The vehicle skids 3 cm after braking, and the egg tip sits 10 cm ahead of the point used as the start reference. How many turns?
Separate investigation observations:
Setting 30.0 turns: Stops at 7.21, 7.24, 7.19 m.
Setting 35.0 turns: Stops at 8.40, 8.43, 8.38 m.
Target: 8.00 m.

A. A remote control

B. A tether from the start line

C. A pre-loaded rubber band that only operates the brake

D. A battery-powered motor

**Correct: C**

**4. (2 points) Why might a team use less than 2.00 kg?**

A. It increases distance

B. Heavier masses are slower

C. Less energy makes stopping gentler and protects the egg

D. The rules require less

**Correct: C**

**5. (2 points) The vehicle does not move when triggered. What happens?**

Unit: Calibration and competition day.
Worked-case data: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.
Separate investigation observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

A. A run has not occurred; the team may keep working

B. It is a Failed Run

C. It scores 2500

D. The team is disqualified

**Correct: A**

**6. (4 points) Why is the egg required to be foremost?**

Unit: Protecting the egg.
Worked-case data: A vehicle arrives at 1.5 m/s and stops in 6 cm. Find the average deceleration in g.
Separate investigation observations:
Design A: Arrives 3.0 m/s, stops in 2 cm; 2 of 5 eggs cracked.
Design B: Arrives 1.5 m/s, stops in 5 cm; 0 of 5 cracked.
Question: Compare decelerations.

answer: Vehicle Distance is measured from the egg tip, and the egg leads the vehicle to the barrier. (4 points).

**7. (4 points) How can a team check alignment quickly?**

Unit: Drivetrains: turning a drop into distance.
Worked-case data: A vehicle has 0.50 cm axle radius and 4.0 cm wheel radius. The mass pulls 0.75 m of string. About how far can the vehicle be driven?
Separate investigation observations:
Run 1: String wound in one neat layer: 8.42 m.
Run 2: String piled in three layers: 7.61 m.
Same: Mass and drop height.

answer: Push the vehicle gently and watch whether it rolls straight and far without scrubbing. (4 points).

**8. (4 points) Why must the falling mass not hit the floor?**

Unit: Rules and the low-score system.
Worked-case data: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations. Find the Run Score.
Separate investigation observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

answer: The rules forbid floor contact to prevent damage, so teams must use a cushion within the size limits. (4 points).

**9. (4 points) How can a vehicle be made to follow a gentle arc?**

Unit: The Can Bonus and curved paths.
Worked-case data: Target 9.00 m. The team sets a 60 cm gap. Find the bonus, the gap center offset, and the arc radius.
Separate investigation observations:
Arc runs: Passed between cans 10 of 10.
Stopping: Stopped 18 cm short of the target each time.
Setting: Turns calculated from the straight-line distance.

answer: Angle the wheels slightly or use slightly different drive wheel sizes so one side travels farther. (4 points).

**10. (4 points) Why measure Vehicle Distance from the egg tip?**

Unit: Automatic stopping.
Worked-case data: Target 9.25 m. Wheels are 7.50 cm in diameter. The vehicle skids 3 cm after braking, and the egg tip sits 10 cm ahead of the point used as the start reference. How many turns?
Separate investigation observations:
Setting 30.0 turns: Stops at 7.21, 7.24, 7.19 m.
Setting 35.0 turns: Stops at 8.40, 8.43, 8.38 m.
Target: 8.00 m.

answer: The rules define Vehicle Distance from the egg tip, the Vehicle Measurement Point, to the Target Point. (4 points).

**11. (6 points) Solve this case and explain your reasoning: A 1.50 kg mass falls 0.60 m. If 40 percent reaches a 0.50 kg vehicle as kinetic energy, what is its top speed?**

reasoning: Energy released = 1.50 x 9.8 x 0.60 about 8.8 J.
To vehicle: 0.40 x 8.8 about 3.5 J.
v = sqrt (2E ÷ m) = sqrt (2 x 3.5 ÷ 0.50) about 3.7 m/s. (3 points).

conclusion: About 3.7 m/s, fast enough that braking must be gentle. (3 points).

**12. (6 points) Solve this case and explain your reasoning: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.**

reasoning: Slope: (9.10 - 7.20) ÷ 8 = 0.2375 m per turn.
From 34 turns: (8.60 - 8.15) ÷ 0.2375 about 1.9 turns.
Set about 35.9 turns. (3 points).

conclusion: About 35.9 turns for 8.60 m. (3 points).

**13. (2 points) Why did Design B protect the eggs?**

Supplied synthetic observations:
Design A: Arrives 3.0 m/s, stops in 2 cm; 2 of 5 eggs cracked.
Design B: Arrives 1.5 m/s, stops in 5 cm; 0 of 5 cracked.
Question: Compare decelerations.

A. Its deceleration was about 10 times lower

B. It was heavier

C. It traveled farther

**Correct: A**

**14. (2 points) Define “Drive axle” in the context of Scrambler.**

answer: Axle turned by the string. (2 points).

**15. (2 points) Define “Terminal Barrier” in the context of Scrambler.**

answer: Hard wall at least 25 cm tall centered at the Target Point. (2 points).

**16. (2 points) Define “Outside Can” in the context of Scrambler.**

answer: Placed by the supervisor, inside edge 100 cm from center. (2 points).

**17. (2 points) Define “Wing nut” in the context of Scrambler.**

answer: Nut that travels along the threads to trigger the brake. (2 points).

**18. (2 points) Define “Rolling resistance” in the context of Scrambler.**

answer: Energy loss as wheels roll. (2 points).